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United States Patent	12396751
Kind Code	B2
Date of Patent	August 26, 2025
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Nested cannula system

Abstract

The present disclosure provides for an interlocking cannula system that includes a handle and a set of cannulas. The set of cannulas includes a sequence of smaller to larger cannulas constructed such that the smaller cannulas may be nested within larger cannulas. Each cannula in the set of cannulas includes a respective locking mechanism configured such that each cannula may be interlocked with each of the other cannulas in the set. The handle is constructed such that each cannula in the set may interlock with the handle by the respective locking mechanism of the cannula. In some instances, the respective locking mechanisms are constructed to enable an ordered release of the cannulas.

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Appl. No.: 18/531382

Filed: December 06, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240099742 A1	Mar. 28, 2024

Related U.S. Application Data

continuation parent-doc US 17153391 20210120 US 11857217 child-doc US 18531382
us-provisional-application US 62964450 20200122

Publication Classification

Int. Cl.: A61B17/34 (20060101)

U.S. Cl.:

CPC A61B17/3439 (20130101); A61B2017/347 (20130101)

Field of Classification Search

CPC: A61B (17/3439); A61B (2017/347)

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Background/Summary

PRIORITY CLAIM (1) The present application is a continuation of U.S. patent application Ser. No. 17/153,391 filed on Jan. 20, 2021, which claims priority to and the benefit of U.S. Provisional Application 62/964,450 filed Jan. 22, 2020, which are each incorporated herein by reference in their entirety for all purposes.

BACKGROUND

(1) Various surgical procedures often require surgeons to access many different areas of a patient's body. In some instances, however, human anatomy prevents a surgeon from direct access to the subject area of a procedure. For instance, bone or soft tissue may be blocking the surgeon's access, and thus the surgeon must drill a hole through the bone or make an incision in the soft tissue to access the subject area. To maintain access to the subject area, a surgeon may insert a cannula. A cannula includes a hollow tube and thus enables the surgeon to insert various surgical instruments through the hollow tube to access the subject area for the surgical procedure. The cannula may also protect the patient's soft tissue surrounding the cannula from damage by the surgical instruments.

(2) A surgical procedure often requires a surgeon to use multiple different surgical instruments, which are often different sizes. Thus, when a surgeon uses a cannula, the cannula must have a large enough passageway such that the largest instrument (e.g., implant) for the procedure may fit through the cannula. For some instruments, however, if the cannula passageway is too large as compared to the instrument, it may be more difficult for the surgeon to finely control the instrument. Thus, a surgeon may preferably need different sized cannulas for different instruments.

(3) One way for a surgeon to be provided different-sized cannulas is for the surgeon to switch out a

cannula each time the surgeon needs a different size. This method, however, provides the surgeon with extra labors with how to perform a procedure. For instance, in some instances, it may increase surgical procedure time and the potential for soft tissue irritation/damage as changing the size of the cannula necessitates removing the cannula from the surgical site and inserting the desired cannula into the site. Additionally, if the incision was not made large enough to fit the largest required cannula, the surgeon may have to increase the size of the incision mid-surgery, adding to the surgical time. In another instance, an incision made for a large cannula will not be suitable for a small cannula, and thus once a surgeon makes an incision for a large cannula, the surgeon can no longer use a small cannula unless the surgeon makes a new incision at a different location.

Accordingly, the surgeon is limited to using surgical instruments in increasing size of surgical instrument, which is not desirable. It is additionally not desirable for a surgeon to make multiple incisions for different-sized cannulas as this increases a patient's healing time.

(4) Another way for a surgeon to be provided different-sized cannulas is a nested cannula system. A nested cannula system includes multiple different-sized cannulas that may nest within one another in a telescoping fashion. Thus, when a surgeon needs a smaller cannula, the surgeon may insert the smaller cannula within the larger cannula that is already within the patient, rather than switching the cannulas or making a new incision. Conventional nested cannula systems, however, may include an attached handle on a single cannula size, thus necessitating that a surgeon always use the cannula size with the attached handle. This provides the surgeon with limited flexibility, and in some instances, may require the surgeon to use a larger cannula than needed, which requires the surgeon to make a larger incision (or drill a larger bone hole) than would be required with a smaller cannula. The larger incision or bone hole results in a longer recovery time for the patient.

(5) Some conventional nested cannula systems may include attached handles on multiple cannula sizes. The additional handles in such systems, however, generate additional weight that may place additional stress on a patient's soft tissue, bone, joint, etc. The additional handles also generate additional components that a surgeon must be mindful of and control when performing a procedure.

(6) In addition, some conventional nested cannula systems may rely on the diametric fit between cannula sizes for maintaining concentricity between the smallest and largest cannulas. This reliance accordingly necessitates that a surgeon use all intermediate cannulas between two sizes, which also generates additional weight that may place additional stress on a patient's soft tissue, bone, joint, etc. Some conventional nested cannula systems additionally rely on friction between adjacently-sized cannulas to retain one cannula with another. This reliance also necessitates that a surgeon use all intermediate cannulas between two sizes. Additionally, friction-based retention may disengage and/or allow rotation between cannulas when forces are applied through a surgical instrument to the underlying bone in contact with the outermost cannula.

(7) Accordingly, a need exists for a nested cannula system that provides surgeons with greater flexibility and reduces patient recovery times as compared to conventional nested cannula systems.

SUMMARY

(8) The present disclosure provides new and innovative interlocking nested cannula systems and methods for surgical procedures. Each cannula in the provided system may be used individually with a handle, or may be interlocked with any other cannula in the system. In this way, the provided nested cannula system enables a surgeon to use the smallest cannula that is needed for a given procedure and to skip unneeded cannula sizes when switching from one size to another.

(9) In an example, a system includes a handle and a plurality of cannulas. The plurality of cannulas include a sequence of smaller to larger cannulas that are configured such that smaller cannulas may be nested within larger cannulas. Each cannula of the plurality of cannulas includes a respective locking mechanism configured such that each cannula may be interlocked with each of the other cannulas of the plurality of cannulas. Additionally, the handle is configured such that each cannula of the plurality of cannulas may interlock with the handle by the respective locking mechanism of

the cannula.

(10) In another example, a method includes inserting into a patient at least a portion of a set of nested cannulas including a first cannula, a second cannula, and a third cannula. The second cannula is nested within, and interlocked with, the first cannula, and the third cannula is nested within, and interlocked with, the second cannula. Either the third cannula is removed from within the second cannula while the second cannula remains nested within, and interlocked with, the first cannula or the second cannula is removed from within the first cannula while the third cannula remains nested within, and interlocked with, the second cannula.

(11) Additional features and advantages of the disclosed method and apparatus are described in, and will be apparent from, the following Detailed Description and the Figures. The features and advantages described herein are not all-inclusive and, in particular, many additional features and advantages will be apparent to one of ordinary skill in the art in view of the figures and description. Moreover, it should be noted that the language used in the specification has been principally selected for readability and instructional purposes, and not to limit the scope of the inventive subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a perspective view of an example cannula system, according to an aspect of the present disclosure.
- (2) FIGS. 2A and 2B illustrate top views of an example engagement of a locking mechanism, according to an aspect of the present disclosure.
- (3) FIG. 3A illustrates a perspective view of an example configuration of nested cannulas, according to an aspect of the present disclosure.
- (4) FIG. 3B illustrates a perspective view an example configuration of the nested cannulas of FIG. 3A, according to an aspect of the present disclosure.
- (5) FIG. 4A illustrates a perspective view of an alternative locking mechanism, according to an aspect of the present disclosure.
- (6) FIG. 4B illustrates a perspective view of a handle corresponding to the alternative locking mechanism of FIG. 4A, according to an aspect of the present disclosure.
- (7) FIG. 5 illustrates an example procedure using a cannula system, according to an aspect of the present disclosure.
- (8) FIG. 6 illustrates a perspective view of a cannula system including cannulas having respective twist locking mechanisms constructed for ordered release, according to an aspect of the present disclosure.
- (9) FIG. 7 illustrates a side view of the cannulas of FIG. 6, according to an aspect of the present disclosure.
- (10) FIG. 8 illustrates a bottom view of the cannulas of FIG. 6, according to an aspect of the present disclosure.
- (11) FIG. 9 illustrates a top view of the cannulas of FIG. 6, according to an aspect of the present disclosure.
- (12) FIG. 10 illustrates a perspective view of a cannula system including cannulas having respective spring-and-thread locking mechanisms constructed for ordered release, according to an aspect of the present disclosure.
- (13) FIG. 11 illustrates a side view of the cannulas of FIG. 10, according to an aspect of the present disclosure.
- (14) FIG. 12 illustrates a bottom view of the cannulas of FIG. 10, according to an aspect of the present disclosure.

(15) FIG. 13 illustrates a top view of the cannulas of FIG. 10, according to an aspect of the present disclosure.

DETAILED DESCRIPTION

(16) The present disclosure relates generally to cannulas for surgical procedures. More specifically, the present disclosure provides a nested cannula system in which each cannula in the system may be used individually with a handle, or may be interlocked with any other cannula in the system. When two cannulas are interlocked with one another, one cannula is nested within a second cannula. The handle is a separate handle component of the provided cannula system that may be used with each cannula individually. In this way, the presently disclosed nested cannula system enables a surgeon to use the smallest cannula that is needed for a given procedure, which results in smaller incisions or bone holes and faster patient recovery times. The presently disclosed nested cannula system additionally enables a surgeon to skip unneeded cannula sizes when switching from one size to another since each cannula may be used with any other cannula by way of respective locking mechanisms on each cannula. In this way, the provided cannula system can help place less stress on a patient's soft tissue, bone, joint, etc. by eliminating the need to use all intermediate cannulas as compared to typical nested cannula systems.

(17) The respective locking mechanisms of the provided cannulas interlock two nested cannulas with one another. In at least some aspects, the respective locking mechanisms maintain concentricity between two nested cannulas. In some aspects, a respective locking mechanism may prevent rotation between two nested cannulas with respect to one another. In some aspects, the respective locking mechanisms may be constructed to enable an ordered release of nested and interlocked cannulas such that, for example, it is easier (e.g., requires less force) to disengage and remove a first cannula from a second cannula than it is to disengage and remove the second cannula from a third cannula. The ordered release can help a surgeon disengage and remove the first cannula without disengaging or removing the second and third cannulas, and in at least some instances, can help the surgeon do this with one hand to free up the surgeon's other hand during surgery.

(18) FIG. 1 illustrates an example cannula system **100**, according to an aspect of the present disclosure. In some aspects, the example cannula system **100** includes a handle **120** and three differently-sized cannulas **110A**, **110B**, and **110C**. In other aspects, the cannula system **100** may include more than three differently-sized cannulas **110A**, **110B**, **110C**. The cannula **110A** includes a head **116A**, and tube **112A** with a diameter **114A**. The cannula **110B** similarly includes a head **116B**, and a tube **112B** with a diameter **114B**. The cannula **110C** similarly includes a head **116C**, and a tube **112C** with a diameter **114C**.

(19) The cannulas **110A**, **110B**, and **110C** are sized in a sequence of smaller cannulas to larger cannulas. For instance, the cannula **110C** is larger than the cannula **110B**, which is larger than the cannula **110A**. For example, the diameter **114C** of the tube **112C** is larger than the diameter **114B** of the tube **112B**, which is larger than the diameter **114A** of the tube **112A**. In aspects in which the cannula system **100** includes more than three cannulas **110A**, **110B**, **110C**, the sequence may further include a cannula larger than the cannula **110C** or a cannula smaller than the cannula **110A**, and so forth.

(20) In some aspects of the present disclosure, it may be only the respective tubes **112A**, **112B**, and **112C** of each cannula **110A**, **110B**, **110C** in the cannula system **100** that change in size. Stated differently, in such aspects only the diameters **114A**, **114B**, and **114C** of the respective tubes **112A**, **112B**, and **112C** change in size among the sequence of cannulas while the size of the respective heads **116A**, **116B**, and **116C** remain the same. Each cannula **110A**, **110B**, **110C** has a channel **120A**, **120B**, **120C** that extends the length of the cannula **110A**, **110B**, **110C**, through the respective heads **116A**, **116B**, **116C** and respective tubes **112A**, **112B**, **112C**, with an opening at each end. The difference in diameters **114A**, **114B**, **114C** between the tubes **112A**, **112B**, **112C** of the cannulas **110A**, **110B**, **110C** enables the cannulas **110A**, **110B**, and **110C** to be nested within one another. For

example, the tube **112A** of the cannula **110A** may be placed within the channel **120B** of the cannula **110B** or the channel **120C** of the cannula **110C**, and the tube **112B** of the cannula **110B** may be placed within the channel **120C** of the cannula **110C**.

(21) In various aspects of the present disclosure, the example handle **120** of the cannula system **100** includes a gripping portion **124** and an opening **122**. In some aspects, the gripping portion **124** may have suitable shapes other than that illustrated and/or may include additional material for enhancing grip, such as rubber, on the outer surface. The opening **122** of the handle **120** is configured according to a locking mechanism of each cannula **110A**, **110B**, **110C** in order to enable interlocking a respective cannula **110A**, **110B**, **110C** with the handle **120**. For example, in some aspects, such as the one illustrated, each locking mechanism is a twist lock mechanism such that a first cannula (e.g., the cannula **110A**) is twisted or rotated relative to a second cannula (e.g., the cannula **110B**) in order to lock the first and second cannulas. In such an example, the twist lock mechanism may include a male lock component and a female lock component or opening. The female lock opening allows the male lock component to pass through when the male lock component is in one orientation, but does not let it pass through when the male lock component is rotated relative to the female lock opening.

(22) In an example, each cannula **110A**, **110B**, **110C** includes a male lock component **118A**, **118B**, **118C** that includes a cross-sectional shape with two straight sides and two curved sides. Each cannula **110A**, **110B**, **110C** also includes a female lock component or opening (not illustrated), which will be described in more detail below. In various aspects, the example opening **122** of the handle **120** has the same corresponding cross-sectional shape as each male lock component **118A**, **118B**, and **118C**. In such aspects, when the male lock component **118A** of the cannula **110A** is oriented to match its straight and curved sides with the opening **122**, the male lock component **118A** may pass through the opening **122**. However, if the cannula **110A** is rotated once the male lock component **118A** is through the opening **122**, the straight and curved sides no longer match up, and the male lock component **118A** cannot pass back through the opening **122**. The cannula **110A** is accordingly interlocked with the handle **120**.

(23) FIGS. 2A and 2B illustrate a top view of an example cannula **210** placed through an opening in the example handle **200** in an unlocked and locked position, respectively, according to an aspect of the present disclosure. In FIG. 2A, the cross-section of the male lock component of the example cannula **210** matches the opening in the handle **200**. However, in FIG. 2B, the example cannula **210** is rotated and the cross-section of the male lock component no longer matches the opening in the handle **200**. The male lock component of the example cannula **210**, therefore, may no longer pass through the opening and the cannula **210** is interlocked with the handle **200**.

(24) FIGS. 2A and 2B additionally illustrate a top view of an example female lock component or opening **220** of the example cannula **210**. In various aspects, the example female lock opening **220** is similar to the opening of the handle **200** in that it has a cross-sectional shape that matches the male lock component in one orientation, but not when the male lock component is rotated relative to the female lock opening **220**. In some aspects, the female lock opening **220** includes a notch in the inner wall of the female lock opening **220** that is configured to accept the male lock component.

(25) Each cannula **110A**, **110B**, **110C** in the cannula system **100** can be interlocked with any other cannula **110A**, **110B**, **110C** in the cannula system **100**. For example, FIGS. 3A and 3B illustrate two example orientations of the cannulas **110A**, **110B**, and **110C**. In the example orientation of FIG. 3A, the cannula **110C** is interlocked with the handle **120**. The cannula **110B** is nested within, and interlocked with, the cannula **110C**. And, the cannula **110A** is nested within, and interlocked with, the cannula **110B**. In comparison, the example orientation of FIG. 3B includes the cannula **110C** interlocked with the handle **120**; however, the cannula **110A** is nested within, and interlocked with, the cannula **110C**, while the cannula **110B** remains unused and off to the side. In other examples, the cannula **110B** or the cannula **110A** may be interlocked with the handle **120** and the cannula **110C** may be unused.

(26) The ability to interlock any of the cannulas **110A**, **110B**, **110C** with each other or with the handle **120** provides a surgeon with greater flexibility in choosing which cannulas to use for a surgical procedure than conventional cannula systems. This increased flexibility may enable a surgeon to make smaller incisions in a patient since the surgeon is not required to use the largest cannula in a set. The increased flexibility may also lessen the weight of, and thus the force applied on the patient by, the nested cannulas since the surgeon may skip cannula sizes when nesting one cannula within another.

(27) In some aspects of the present disclosure, the respective locking mechanisms of each cannula **110A**, **110B**, **110C** may be identical or substantially identical. Stated differently, in such aspects, each male lock component **118A**, **118B**, **118C** of each respective cannula **110A**, **110B**, **110C** in the cannula system **100** are substantially identical, and each female lock opening **220** of each respective cannula **110A**, **110B**, **110C** in the cannula system **100** are substantially identical. In an example, each male lock component **118A**, **118B**, **118C** and each female lock opening **220** are sized equally regardless of the size of their respective cannula **110A**, **110B**, **110C**. In such aspects, the cannulas **110A**, **110B**, and **110C** can be considered to have a universal locking mechanism since the locking mechanism of each is the same.

(28) In other aspects, the cannulas **110A**, **110B**, and **110C** could have identical or substantially identical male lock components **118A**, **118B**, and **118C**, but different female lock openings **220**. In other aspects still, the cannulas **110A**, **110B**, and **110C** could have different male lock components **118A**, **118B**, and **118C**, but identical or substantially identical female lock openings **220**. In some aspects, the smallest cannula (e.g., the cannula **110A**) in the system **100** does not have a female lock opening **220** since no other cannula (e.g., the cannulas **110B** and **110C**) can nest within the smallest cannula.

(29) In some aspects of the present disclosure, a respective locking mechanism of each cannula of the cannula system **100** may be a squeeze lock mechanism. FIG. 4A shows example cannulas **410A** and **410B** configured with an example squeeze lock mechanism. For instance, the example cannula **410A** includes gaps **414A** and **414A'** that extend through the head **416A** of the example cannula **410A** into a portion of the tube **418** of the example cannula **410A**. The gaps **414A** and **414A'** enable the cannula **410A** to be compressed inward by applying a force on either side of the cannula **410** to reduce the size of the gaps **414A** and **414A'**. For example, a force may be applied to the head **416A** of the cannula **410A** in the direction of the arrows illustrated on the example cannula **410B**. The example cannula **410A** may also include an extension **412A**.

(30) Accordingly, the combination of the ability to be flexed inward and the extension **412A** enable the example cannula **410A** to be interlocked with the handle **400**. For instance, when the cannula **410A** is compressed inward, the extension **412A** may be able to pass through the opening **420** (FIG. 4B) of the handle **410**. However, when the compressive force is released after the extension **412A** passes through the opening **420**, the extension **412A** extends past the outer bounds of the opening **420** and may not pass back through the opening **420**. Thus, this enables the handle **400** to be interlocked with the cannula **410**.

(31) Due to the gaps **412A** and **412A'**, the extension **412A** includes multiple extension components that are broken up by the gaps **412A** and **412A'**. In some examples, the extension **412A** may extend fully around the perimeter of the cannula **410A**, except for the gaps **412A** and **412A'**. In some examples, the extension **412A** may have a circular cross-section corresponding to the circular opening **420** of the handle **400** as illustrated in FIG. 4B. In such examples, the cannula **410A** may rotate freely about its axis in either direction since the opening **420** is circular. In other examples, the extension **412A** may have other cross-sectional shapes that correspond to the cross-sectional shape of the opening **420** of the handle **400**, for instance, a square. In such an instance, the cannula **410A** may include an extension **412A** only on one or more sides of the square, such as on two sides. In such an instance, the cannula **410A** is prevented from rotating about its axis since the opening **420** is square.

(32) Similar to how the example cannula **410A** may be interlocked with the handle **400**, two cannulas (e.g., the cannulas **410A** and **410B**) may be interlocked with one another by way of the squeeze lock mechanism. In such instances, the extension **412A** may be considered a male lock component and the head **416A** of the cannula **410A** may have a female lock component, such as an opening and a notch. For example, the example cannula **410B** similarly includes gaps **414B** and **414B'** and an extension **412B** consistent with the description above for the cannula **410A**. Accordingly, the cannula **410B** may be compressed inward by a force in the direction of the illustrated arrows on the head **416B** of the cannula **410B**. The compressive force decreases the gaps **414B** and **414B'** and thus allows the extension **412B** to pass through the opening of the female lock component in the head **416A** of the cannula **410A**. Once the extension **412B** passes through the opening, the compressive force may be released, increasing the gaps **414B** and **414B'** and allowing the extension **412B** to enter a notch within the inner wall of the head **416A** of the cannula **410A**. When the extension **412B** is within the notch, it may not pass back through the opening, and thus the cannula **410B** is interlocked with the cannula **410A**.

(33) In some examples, the notch of the female lock component in the head **416A** of the cannula **410A** may extend around the entire perimeter of the inner wall of the head **416A**. Such examples may enable the cannula **410B** to rotate freely about its axis when it is nested within, and interlocked with, the cannula **410A**. For instance, in some examples, the cross-section of the notch and of the extension **412B** may be correspondingly circular. In some examples, the notch may extend less than the entire perimeter of the inner wall. Such examples may prevent or limit the cannula **410B** from rotating about its axis when it is nested within, and interlocked with, the cannula **410A**. For instance, the cannula **410A** may include multiple notches that correspondingly match the multiple extension components of the extension **412B** on the cannula **410B**. Thus, when the extension components are placed within their respective notches, the extension components are locked within the notch and the cannula **410B** is prevented from rotating. In some examples, each notch of the multiple notches may be sized larger than each extension component, although the notches are separate and do not join together. Thus, in such examples, each extension component has space to move within its respective notch and the cannula **410B** may partially rotate about its axis in either direction, though it may not freely rotate a complete revolution.

(34) In other examples, the opening and notch of the example cannula **410A** may have other cross-sectional shapes that correspond to the cross-sectional shape of the extension **412B** of the cannula **410B**, for instance, a square. In such an instance, the cannula **410B** may include an extension **412B** only on one or more sides of the square, such as on two sides. In such an instance, the cannula **410B** is prevented from rotating about its axis when it is nested within, and interlocked with, the cannula **410A** because of the square cross-sectional shape.

(35) Each cannula **410A**, **410B** may be interlocked with any other cannula **410A**, **410B** in an example cannula system **100** including respective squeeze lock mechanisms. In some aspects, each respective squeeze lock mechanism on each cannula **410A**, **410B** may be sized equally.

(36) In some aspects of the present disclosure, the respective locking mechanisms of the each cannula **110A**, **110B**, **110C** may be configured so as to enable an ordered release or disengagement of the nested and interlocked cannulas **110A**, **110B**, **110C** from one another. The advantages of such aspects may be exemplified by the following situation. A surgeon may insert the cannulas **110A**, **110B**, and **110C** in the orientation of FIG. 3A at least partially into a patient. The surgeon may later desire to only disengage and remove the cannula **110A** from the cannula **110B**, while the cannula **110B** remains nested within and interlocked with the cannula **110C** and the cannula **110C** remains interlocked with the handle **120**. In some aspects, a surgeon may need to immobilize the cannula **110B** (e.g., hold it with one hand) while disengaging or removing (e.g., with the surgeon's other hand) the cannula **110A** so that the cannula **110B** does not also disengage from the cannula **110C** and the cannula **110C** does not disengage from the handle **120**.

(37) It would be advantageous, however, if the surgeon were able to disengage the cannula **110A**

using only one hand to free the surgeon's other hand during surgery. Accordingly, aspects with cannulas **110A**, **110B**, **110C** that enable an ordered release include respective locking mechanisms that require different amounts of force to disengage the various combinations of nesting orientations of the cannulas **110A**, **110B**, and **110C** and the handle **120**. For example, in the orientation of FIG. 3A, it may require less force to disengage the cannula **110A** from the cannula **110B** than it does to disengage the cannula **110B** from the cannula **110C**. And it requires less force to disengage cannulas **110A**, **110B**, or **110C** from one another than it does to disengage a cannula **110B**, **110B**, or **110C** from the handle **120**. In this way, the surgeon can disengage and remove the cannula **110A** from the cannula **110B** without having to immobilize the cannula **110B**, and without disengaging the cannula **110B** from the cannula **110C** or the cannula **110C** from the handle **120**, by applying a force (e.g., twist or pull) to the cannula **110A** that is sufficient to disengage the cannula **110A** from the cannula **110B** but insufficient to disengage the cannula **110B** from the cannula **110C** or the cannula **110C** from the handle **120**. The surgeon may apply such force using only one hand.

(38) FIGS. 6 to 9 illustrate views of an example cannula system **600** including cannulas **602A**, **602B**, and **602C** having respective twist lock mechanisms constructed for ordered release. Referring to FIG. 6, in various aspects, the example cannula system **600** includes a cannula **602A**, a cannula **602B**, and a cannula **602C**, though in other aspects the cannula system **600** may include additional cannulas, as described similarly for the cannula system **100** discussed above. In at least some aspects, the system **600** includes a handle **120**. The handle **120** is described above. The cannulas **602A**, **602B**, and **602C** are sized in a sequence of smaller cannulas to larger cannulas. For instance, the cannula **602C** includes a tube **112C** which has a larger diameter than the tube **112B** of the cannula **602B**, which has a larger diameter than the tube **112A** of the cannula **602A**. In aspects in which the cannula system **600** includes more than three cannulas **602A**, **602B**, **602C**, the sequence may further include a cannula larger than the cannula **602C** or a cannula smaller than the cannula **602A**, and so forth.

(39) In at least some aspects, the cannula **602A** includes a male lock component having a thread **606A** and a cam **608**. The male lock component of the cannula **602A** may be integral with or attached to a head **604A** of the cannula **602A**. In some examples, the head **604A** may include teeth or another suitable grip-enhancing construction or material. In at least some aspects, the thread **606A** is split into two portions. For example, the thread **606A** may be split by opposing flat surfaces **612A** and **802A** (FIG. 8). FIG. 7 illustrates the cannula **602A** rotated ninety degrees relative to the illustrated cannula **602A** in FIG. 6 to show the thread pitch of the thread **606A**. FIG. 8 illustrates a bottom view of the cannula **602A**. In at least some aspects, the cam **608** may have the shape illustrated in FIG. 8, though in other aspects may have other suitable shapes. FIG. 9 illustrates a top view of the cannula **602A** showing an opening to a channel **902A** that extends through the tube **112A** of the cannula **602A**. In this example, the cannula **602A** is the smallest cannula in the system **600** and does not include a female lock component or opening.

(40) Returning to FIG. 6, in various aspects, the cannula **602B** includes a male lock component having a thread **606B**. In at least some examples, the male lock component of the cannula **602B** does not include a cam. In some aspects, the male lock component of the cannula **602B** instead includes a support member **610**. In such aspects, the support member **610** can help in maintaining concentricity of the cannula **602B** with the cannula **602C** when nested within the cannula **602C**. FIG. 8 illustrates a bottom view of the cannula **602B**. In some aspects, such as the illustrated aspect, the support member **610** may have a circular shape.

(41) Returning to FIG. 6, the male lock component of the cannula **602B** may be integral with or attached to a head **604B** of the cannula **602B**. In some examples, the head **604B** may include teeth or another suitable grip-enhancing construction or material. In at least some aspects, the thread **606B** is split into two portions. For example, the thread **606B** may be split by opposing flat surfaces **612B** and **802B** (FIG. 8). FIG. 7 illustrates the cannula **602B** rotated ninety degrees relative to the illustrated cannula **602B** in FIG. 6 to show the thread pitch of the thread **606B**. In at

least some aspects, the thread pitch of the thread **606B** is equal to the thread pitch of the thread **606A** of the cannula **602A**. FIG. 9 illustrates a top view of the cannula **602B** showing an opening to a channel **902B** that extends through the tube **112B** of the cannula **602B**. The cannula **602B** includes a female lock component/opening having a notch **904B**. The notch **904B** has a shape that suitably corresponds to the cam **608** of the cannula **602A**. In some aspects, the female lock component/opening of the cannula **602B** includes a female thread (not shown) that extends into the sidewalls **906B** and **908B** of the head **604B**.

(42) Returning to FIG. 6, in various aspects, the cannula **602C** includes a male lock component having a thread **606C**. The male lock component of the cannula **602C** may be integral with or attached to a head **604C** of the cannula **602C**. In some examples, the head **604C** may include teeth or another suitable grip-enhancing construction or material. In at least some aspects, the thread **606C** is split into two portions. For example, the thread **606C** may be split by opposing flat surfaces **612C** and **802C** (FIG. 8). FIG. 7 illustrates the cannula **602C** rotated ninety degrees relative to the illustrated cannula **602C** in FIG. 6 to show the thread pitch of the thread **606C**. In at least some aspects, the thread **606C** has a smaller thread pitch than the thread **606A** of the cannula **602A** and the thread **606B** of the cannula **602B**. FIG. 8 illustrates a bottom view of the cannula **602C**.

(43) FIG. 9 illustrates a top view of the cannula **602C** showing an opening to a channel **902C** that extends through the tube **112C** of the cannula **602C**. The cannula **602C** includes a female lock component/opening. In at least some aspects, the female lock component/opening of the cannula **602C** includes a notch **904B**. In such aspects, the notch **904B** is sized and shaped such that it does not suitably correspond to the size and shape of the cam **608** of the cannula **602A**. The notch **904B** may, in some instances, be sized and shaped to correspond to the support member **610** of the cannula **602B**, which may help the main concentricity of the cannula **602B** and the cannula **602C** when the cannula **602B** is nested within the cannula **602C**. In at least some aspects, the female lock component/opening of the cannula **602C** includes a female thread (not shown) that extends into the sidewalls **906C** and **908C** of the head **604C**.

(44) The interaction of the cannulas **602A**, **602B**, and **602C** will now be described. The cam **608** of the male lock component of the cannula **602A** can engage with the sidewall of the notch **904B** of the female lock component/opening of the cannula **602B**, for instance by rotating the cannula **602A** when nested within the cannula **602B**. The engagement of the cam **608** with the sidewall of the notch **904B** interlocks the cannula **602A** with the cannula **602B**. The cam **608**, however, will not engage with the sidewall of the notch **904C** of the female lock component/opening of the cannula **602C**.

(45) The thread **606A** of the male lock component of the cannula **602A** does not engage the female thread extending into the sidewalls **906B** and **908B** of the female lock component/opening of the cannula **602B**. Rather, the thread **606A** merely fits within the female thread of the cannula **602B**. For example, if the cannula **602A** is rotated just past the cam **608** becoming disengaged from the sidewall of the notch **904B**, the female thread of the cannula **602B** will prevent a surgeon from translating the cannula **602A** along the long axis of the cannulas **602A** and **602B** to remove the cannula **602A** from the cannula **602B**; however, the cannula **602A** will be loose to translate back and forth along that long axis because the thread **606A** is not engaged with the female thread of the cannula **602B**. Once the cannula **602A** is rotated further, the thread **606A** becomes clear of the female thread of the cannula **602B** and the surgeon may remove the cannula **602A** from the cannula **602B**. In this way, while the female thread of the cannula **602B** does not contribute to the engagement of the cannula **602A** with the cannula **602B**, and therefore does not contribute to the torque required to disengage the cannula **602A** from the cannula **602B**, the female thread of the cannula **602B** can help maintain the connection between the cannula **602A** and the cannula **602B** during surgery if the cam **608** were to accidentally become disengaged from the sidewall of the notch **904B**.

(46) Conversely, the thread **606A** of the cannula **602A** does engage the female thread of the female lock component/opening of the cannula **602C**. The engagement of the thread **606A** with the female thread interlocks the cannula **602A** with the cannula **602C**. As stated above, the cam **608** of the cannula **602A** does not engage with the female lock component/opening of the cannula **602C**. The thread **606A** of the cannula **602A** is also able to interlock with the handle **120** at the opening **122**.

(47) In addition, the thread **606B** of the male lock component of the cannula **602B** engages the female thread of the female lock component/opening of the cannula **602C**. The engagement of the thread **606B** with the female thread interlocks the cannula **602B** with the cannula **602C**. The thread **606B** is also able to interlock with the handle **120** at the opening **122**. The thread **606C** of the male lock component of the cannula **602C** is additionally able to interlock with the handle **120** at the opening **122**.

(48) The above-described respective male lock components and female lock components/openings of the cannulas **602A**, **602B**, and **602C** enable for an ordered release or disengagement of the cannulas **602A**, **602B**, and **602C**. In an example, the cannula **602A** is nested within, and interlocked with the cannula **602B**, which is nested within, and interlocked with the cannula **602C**, which is interlocked with the handle **120**. In this example, it requires less rotational force to disengage the cam **608** of the cannula **602A** from the notch **904B** of the female lock component/opening of the cannula **602B** than it does to disengage the thread **606B** of the cannula **602B** from the female thread of the female lock component/opening of the cannula **602C**. It also requires less rotational force to disengage the cam **608** from the notch **904B** than it does to disengage the thread **606C** of the cannula **602C** from the handle **120**. In this way, a surgeon can disengage and remove the cannula **602A** from the cannula **602B** using one hand without disengaging the cannula **602B** or the cannula **602C**.

(49) Continuing with this example with the cannula **602A** removed, it requires less rotational force to disengage the thread **606B** of the cannula **602B** from the female thread of the female lock component/opening of the cannula **602C** than it does to disengage the thread **606C** of the cannula **602C** from the handle **120**. In this way, a surgeon can disengage and remove the cannula **602B** from the cannula **602C** using one hand without disengaging the cannula **602C**. This is true whether or not the cannula **602A** is interlocked with cannula **602B**.

(50) In some instances, the cannula **602A** may be nested within, and interlocked with, the cannula **602C**, which is interlocked with the handle **120**. In such instances, it requires less rotational force to disengage the thread **606A** of the cannula **602A** from the female thread of the female lock component/opening of the cannula **602C** than it does to disengage the thread **606C** of the cannula **602C** from the handle **120**. In this way, a surgeon can disengage and remove the cannula **602A** from the cannula **602C** using one hand without disengaging the cannula **602C**. In some instances, the cannula **602A** may be nested within, and interlocked with, the cannula **602B**, which is interlocked with the handle **120**. In such instances, it requires less rotational force to disengage the cam **608** of the cannula **602A** from the notch **904B** of the female lock component/opening of the cannula **602B** than it does to disengage the thread **606B** of the cannula **602B** from the handle **120**. In this way, a surgeon can disengage and remove the cannula **602A** from the cannula **602B** using one hand without disengaging the cannula **602B**.

(51) FIGS. **10** to **13** illustrate views of an example cannula system **1000** including cannulas **1002A**, **1002B**, and **1002C** having respective spring-and-thread lock mechanisms constructed for ordered release. Referring to FIG. **10**, in various aspects, the example cannula system **1000** includes a cannula **1002A**, a cannula **1002B**, and a cannula **1002C**, though in other aspects the cannula system **1000** may include additional cannulas, as described similarly for the cannula systems **100** and **600** discussed above. In at least some aspects, the system **1000** includes a handle **120**. The handle **120** is described above. The cannulas **1002A**, **1002B**, and **1002C** are sized in a sequence of smaller cannulas to larger cannulas. For instance, the cannula **1002C** includes a tube **112C**, which has a larger diameter than the tube **112B** of the cannula **1002B**, which has a larger diameter than the tube

112A of the cannula **1002A**. In aspects in which the cannula system **1000** includes more than three cannulas **1002A**, **1002B**, **1002C**, the sequence may further include a cannula larger than the cannula **1002C** or a cannula smaller than the cannula **1002A**, and so forth.

(52) In at least some aspects, the cannula **1002A** includes a male lock component having a thread **1006A** and a spring **1008A**. At least a portion of the male lock component of the cannula **1002A** may be integral with or attached to a head **1004A** of the cannula **1002A**. In some examples, the head **1004A** may include teeth or another suitable grip-enhancing construction or material. In at least some aspects, the spring **1008A** is a canted coil spring. In various examples, the spring **1008A** is partially positioned within a notch of the male lock component of the cannula **1002A** such that the spring **1008A** extends beyond a directly adjacent surface of the male lock component and is prevented from axial movement along the cannula **1002A**. In some aspects, the spring **1008A** is attached to the male lock component of the cannula **1002A**. In other aspects, the spring **1008A** is merely maintained within the notch of the male lock component of the cannula **1002A** and is not otherwise attached to the male lock component.

(53) In at least some aspects, the thread **1006A** of the male lock component of the cannula **1002A** is split into two portions. For example, the thread **1006A** may be split by opposing flat surfaces **1012A** and **1202A** (FIG. 12). FIG. 11 illustrates the cannula **1002A** rotated ninety degrees relative to the illustrated cannula **1002A** in FIG. 10 to show the thread pitch of the thread **1006A**. FIG. 12 illustrates a bottom view of the cannula **1002A** that shows the opposing flat surfaces **1012A** and **1202A**. The bottom view also illustrates that only a portion of the spring **1008A** is visible due to the spring **1008A** being positioned partially within the notch of the male lock component of the cannula **1002A**. FIG. 13 illustrates a top view of the cannula **1002A** showing an opening to a channel **1002A** that extends through the tube **112A** of the cannula **1002A**. In this example, the cannula **1002A** is the smallest cannula in the system **1000** and does not include a female lock component or opening.

(54) Returning to FIG. 10, in at least some aspects, the cannula **1002B** includes a male lock component having a thread **1006B** and a spring **1008B**. At least a portion of the male lock component of the cannula **1002B** may be integral with or attached to a head **1004B** of the cannula **1002B**. In some examples, the head **1004B** may include teeth or another suitable grip-enhancing construction or material. In at least some aspects, the spring **1008B** is a canted coil spring. In various examples, the spring **1008B** is partially positioned within a notch of the male lock component of the cannula **1002B** such that the spring **1008B** extends beyond a directly adjacent surface of the male lock component and is prevented from axial movement along the cannula **1002B**. In some aspects, the spring **1008B** is attached to the male lock component of the cannula **1002B**. In other aspects, the spring **1008B** is merely maintained within the notch of the male lock component of the cannula **1002B** and is not otherwise attached to the male lock component.

(55) In at least some aspects, the thread **1006B** of the male lock component of the cannula **1002B** is split into two portions. For example, the thread **1006B** may be split by opposing flat surfaces **1012B** and **1202B** (FIG. 12). FIG. 11 illustrates the cannula **1002B** rotated ninety degrees relative to the illustrated cannula **1002B** in FIG. 10 to show the thread pitch of the thread **1006B**. FIG. 12 illustrates a bottom view of the cannula **1002B** that shows the opposing flat surfaces **1012B** and **1202B**. The bottom view also illustrates that only a portion of the spring **1008B** is visible due to the spring **1008B** being positioned partially within the notch of the male lock component of the cannula **1002B**. FIG. 13 illustrates a top view of the cannula **1002B** showing an opening to a channel **1002B** that extends through the tube **112B** of the cannula **1002B**. The top view also shows a female lock component/opening of the cannula **1002B**. In at least some aspects, the female lock component/opening of the cannula **1002B** includes a notch **1304B**. The notch **1304B** is sized and shaped in relation to the spring **1008A** of the cannula **1002A**. In at least some aspects, the female lock component/opening of the cannula **1002B** includes a notch **1306B**. The notch **1306B** is sized and shaped to enable space for the thread **1006A** of the cannula **1002A**, though the notch **1306B**

does not engage the thread **1006A**.

(56) Returning to FIG. **10**, in at least some aspects, the cannula **1002C** includes a male lock component having a thread **1006C**. In this example, the cannula **1002C** is the largest cannula in the set of cannulas of the system **1000** and therefore its male lock component does not need to include a spring since the cannula **1002C** cannot nest within any other cannula in the set, though the male lock component of the cannula **1002C** could include a spring. At least a portion of the male lock component of the cannula **1002C** may be integral with or attached to a head **1004C** of the cannula **1002C**. In some examples, the head **1004C** may include teeth or another suitable grip-enhancing construction or material. In at least some aspects,

(57) In at least some aspects, the thread **1006C** of the male lock component of the cannula **1002C** is split into two portions. For example, the thread **1006C** may be split by opposing flat surfaces **1012C** and **1202C** (FIG. **12**). FIG. **11** illustrates the cannula **1002C** rotated ninety degrees relative to the illustrated cannula **1002C** in FIG. **10** to show the thread pitch of the thread **1006C**. In various aspects, the thread **1006A** of the cannula **1002A**, the thread **1006B** of the cannula **1002B**, and the thread **1006C** of the cannula **1002C** have equal thread pitches. FIG. **12** illustrates a bottom view of the cannula **1002C** that shows the opposing flat surfaces **1012C** and **1202C**.

(58) FIG. **13** illustrates a top view of the cannula **1002C** showing an opening to a channel **1002C** that extends through the tube **112C** of the cannula **1002C**. The top view also shows a female lock component/opening of the cannula **1002C**. In at least some aspects, the female lock component/opening of the cannula **1002C** includes a notch **1304C**. The notch **1304C** is sized and shaped in relation to the spring **1008A** of the cannula **1002A** and the spring **1008B** of the cannula **1002B**. Stated differently, the spring **1008A** and the spring **1008B** may be sized and shaped the same so that the notch **1304C** is sized and shaped in relation to each the spring **1008A** and the spring **1008B**. In at least some aspects, the female lock component/opening of the cannula **1002C** includes a notch **1306C**. The notch **1306C** is sized and shaped to enable space for the thread **1006A** of the cannula **1002A** or the thread **1006B** of the cannula **1002B**, though the notch **1306C** does not engage the thread **1006A** nor the thread **1006B**.

(59) The interaction of the cannulas **1002A**, **1002B**, and **1002C** will now be described. The spring **1008A** of the male lock component of the cannula **1002A** can engage with the sidewall of the notch **1304B** of the female lock component/opening of the cannula **1002B**, for instance by translating (e.g., pushing) the cannula **1002A** along an axial direction when nested within the cannula **1002B**. The spring **1008A** is sized larger than the notch **1304B**, and as the spring **1008A** is forced into the notch **1304B**, the spring **1008A** is compressed. The counteracting force of the compressed spring **1008A** against the sidewall of the notch **1304B** interlocks the cannula **1002A** with the cannula **1002B**. To disengage the compressed spring **1008A** from the sidewall of the notch **1304B**, the cannula **1002A** may be translated (e.g., pulled) along the opposite axial direction as when the spring **1008A** was engaged with the sidewall. The spring **1008A** of the male lock component of the cannula **1002A** can engage with the sidewall of the notch **1304C** of the female lock component/opening of the cannula **1002C** in a similar manner, which interlocks the cannula **1002A** with the cannula **1002C**. The thread **1006A** of the male component of the cannula **1002A** is able to interlock (e.g., via rotational engagement) with the handle **120** at the opening **122**.

(60) In addition, the spring **1008B** of the male lock component of the cannula **1002B** can engage with the sidewall of the notch **1304B** of the female lock component/opening of the cannula **1002C**, for instance by translating (e.g., pushing) the cannula **1002B** along an axial direction when nested within the cannula **1002C**. The spring **1008B** is sized larger than the notch **1304C**, and as the spring **1008B** is forced into the notch **1304C**, the spring **1008B** is compressed. The counteracting force of the compressed spring **1008B** against the sidewall of the notch **1304C** interlocks the cannula **1002B** with the cannula **1002C**. To disengage the compressed spring **1008B** from the sidewall of the notch **1304C**, the cannula **1002B** may be translated (e.g., pulled) along the opposite axial direction as when the spring **1008B** was engaged with the sidewall. The thread **1006B** of the male component

of the cannula **1002B** is able to interlock (e.g., via rotational engagement) with the handle **120** at the opening **122**. The thread **1006C** of the male component of the cannula **1002C** is also able to interlock (e.g., via rotational engagement) with the handle **120** at the opening **122**.

(61) The above-described respective male lock components and female lock components/openings of the cannulas **1002A**, **1002B**, and **1002C** enable for an ordered release or disengagement of the cannulas **1002A**, **1002B**, and **1002C**. In an example, the cannula **1002A** is nested within, and interlocked with the cannula **1002B**, which is nested within, and interlocked with the cannula **1002C**, which is interlocked with the handle **120**. In this example, it requires less translational (e.g., pulling) force to disengage the spring **1008A** of the cannula **1002A** from the sidewall of the notch **1304B** of the female lock component/opening of the cannula **1002B** than it does to disengage the spring **1008B** of the cannula **1002B** from the sidewall of the notch **1304C** of the female lock component/opening of the cannula **1002C**. Additionally, since the cannula **1002A** is translated (e.g., pulled) to disengage and remove it from the cannula **1002A**, such translational motion does not apply sufficient rotational force to disengage the thread **1006C** of the cannula **1002C** from the handle **120**. In this way, a surgeon can disengage and remove the cannula **1002A** from the cannula **1002B** using one hand without disengaging the cannula **1002B** or the cannula **1002C**.

(62) Continuing with this example with the cannula **1002A** removed, the spring **1008B** of the cannula **1002B** can be disengaged from the sidewall of the notch **1304C** of the female lock component/opening of the cannula **1002C** by translating (e.g., pulling) the cannula **1002B** along the axial direction. Since the cannula **1002B** is translated (e.g., pulled) to disengage and remove it from the cannula **1002C**, such translational motion does not apply sufficient rotational force to disengage the thread **1006C** of the cannula **1002C** from the handle **120**. In this way, a surgeon can disengage and remove the cannula **1002B** from the cannula **1002C** using one hand without disengaging the cannula **1002C**.

(63) In some instances, the cannula **1002A** may be nested within, and interlocked with, the cannula **1002C**, which is interlocked with the handle **120**. The spring **1008A** of the cannula **1002A** can be disengaged from the sidewall of the notch **1304C** of the female lock component/opening of the cannula **1002C** by translating (e.g., pulling) the cannula **1002A** along the axial direction. Since the cannula **1002A** is translated (e.g., pulled) to disengage and remove it from the cannula **1002C**, such translational motion does not apply sufficient rotational force to disengage the thread **1006C** of the cannula **1002C** from the handle **120**. In this way, a surgeon can disengage and remove the cannula **1002A** from the cannula **1002C** using one hand without disengaging the cannula **1002C**.

(64) In some instances, the cannula **1002A** may be nested within, and interlocked with, the cannula **1002B**, which is interlocked with the handle **120**. The spring **1008A** of the cannula **1002A** can be disengaged from the sidewall of the notch **1304B** of the female lock component/opening of the cannula **1002B** by translating (e.g., pulling) the cannula **1002A** along the axial direction. Since the cannula **1002A** is translated (e.g., pulled) to disengage and remove it from the cannula **1002B**, such translational motion does not apply sufficient rotational force to disengage the thread **1006B** of the cannula **1002B** from the handle **120**. In this way, a surgeon can disengage and remove the cannula **1002A** from the cannula **1002B** using one hand without disengaging the cannula **1002B**.

(65) In some aspects of the present disclosure, each cannula in the set of cannulas of the provided system (e.g., the cannula system **100**, **600**, or **1000**) may be constructed of a particular material to help enable an ordered release of nested and interlocked cannulas. For example, in the example cannula system **600**, the cannula **602A** and the cannula **602B** may be constructed of a heat treated 17-4 stainless steel, the cannula **602C** may be constructed of a Custom 455® heat treated stainless steel, and the handle **120** may be constructed of a commercially pure, grade 4 titanium (titanium CP4). Friction between the different materials in this example helps differentiate the rotational force required to disengage the one cannula from another. For instance, the interface between Custom 455® heat treated stainless steel and titanium CP4 has a greater static friction coefficient than the interface between heat treated 17-4 stainless steel and heat treated 17-4 stainless steel,

which leads to a greater force needed to overcome the friction between Custom 455® heat treated stainless steel and titanium CP4.

(66) FIG. 5 shows an example procedure **500** that includes inserting nested cannulas into a drilled bone hole, according to an aspect of the present disclosure. Alternatively, the procedure **500** may be used external to the bone, protecting the surrounding tissues (e.g., in areas of sensitive soft tissue structures like nerves, where repeated insertion of instrumentation is most likely to cause harm and where surgical duration is important) for the sharp instruments passing through the cannulas.

(67) In other aspects, many other methods of performing the acts described in connection with FIG. 5 may be used. For example, the order of some of the steps may be changed, certain steps may be combined with other steps, one or more of the steps may be repeated, and some of the steps described may be optional. In the example procedure **500**, a surgeon may first determine the cannula sizes that the surgeon will need. In some instances, the surgeon may then nest the cannulas accordingly, if needed, for the size of the first instrument to be used. For example, the surgeon may nest the cannula **530B** within the cannula **530A** and the cannula **530C** within the cannula **530B** so that the size of the cannula **530C** may be used with the first instrument.

(68) In some instances, the surgeon may then drill a hole in the bone **510** corresponding to the largest cannula needed. For example, the cannula **530A** with the tube **532** is illustrated as the largest cannula used in the example procedure **500**. Thus, in this instance, the bone hole will be drilled corresponding to the size of the tube **532**. In some instances, the surgeon may then interlock the handle with largest, outermost cannula to be inserted. For example, the surgeon may interlock the handle **520** with the cannula **530A**. The surgeon may then insert the one or more cannulas through the bone hole. For example, the surgeon may insert the nested cannulas **530A**, **530B**, and **530C** into the hole in the bone **510**.

(69) In some instances, the surgeon may then insert an instrument through the innermost cannula's channel and perform a portion of the example procedure **500**. For example, the surgeon may insert an instrument through the channel of the cannula **530C**. At subsequent points in the example procedure **500**, the surgeon may remove one or more cannulas, or insert one or more cannulas, so that the surgeon may use a larger or smaller instrument that requires a larger or smaller cannula channel. For example, the surgeon may remove the cannula **530C** from the cannula **530B**, for instance by twisting the cannula **530C** to disengage the locking mechanism with cannula **530B**, so that the surgeon may access the channel in the cannula **530B** with a second instrument. At a subsequent time in the example procedure **500**, the surgeon may then remove the cannula **530B** from the cannula **530A**, for instance by twisting the cannula **530B** to disengage the locking mechanism with cannula **530A**, so that the surgeon may access the channel in the cannula **530A** with a third instrument. At a subsequent time in the example procedure **500**, the surgeon may then insert the cannula **530C** into the cannula **530A** and interlock them together, for instance by twisting the cannula **530C** to engage the locking mechanism, so that the surgeon may again use the first instrument.

(70) Without further elaboration, it is believed that one skilled in the art can use the preceding description to utilize the claimed inventions to their fullest extent. The examples and embodiments disclosed herein are to be construed as merely illustrative and not a limitation of the scope of the present disclosure in any way. It will be apparent to those having skill in the art that changes may be made to the details of the above-described embodiments without departing from the underlying principles discussed. In other words, various modifications and improvements of the embodiments specifically disclosed in the description above are within the scope of the appended claims. For example, any suitable combination of features of the various embodiments described is contemplated.

Claims

1. An interlocking cannula system comprising: a plurality of cannulas comprising a first cannula, a second cannula, and a third cannula, wherein the first cannula is smaller than the second cannula and the third cannula, and the second cannula is smaller than the third cannula, thereby allowing the first cannula to be nested within the second cannula and/or the third cannula, wherein each of the plurality of cannulas includes a respective locking mechanism, which allows the first cannula to be interlocked with the third cannula without the second cannula therebetween and which allows the second cannula to be interlocked with the third cannula, wherein the respective locking mechanisms are configured such that interlocking one cannula of the plurality of cannulas with another cannula of the plurality of cannulas includes rotating the one cannula with respect to the other cannula.
2. The interlocking cannula system of claim 1, wherein the plurality of cannulas are configured to allow the first cannula to be removed from the second cannula while the second cannula remains nested within, and interlocked with, the third cannula.
3. The interlocking cannula system of claim 1, wherein the respective locking mechanisms of each of the plurality of cannulas are substantially identical.
4. The interlocking cannula system of claim 1, wherein each respective cannula of the plurality of cannulas includes a head including a first side and a second side, and a tube extending from the second side of the head.
5. The interlocking cannula system of claim 4, wherein the head of each of the cannulas is the same size.
6. The interlocking cannula system of claim 4, wherein the respective locking mechanisms of each of the plurality of cannulas are configured such that the tubes of two interlocked cannulas among the plurality of cannulas are concentric with one another.
7. The interlocking cannula system of claim 4, wherein the head of each of the cannulas includes the respective locking mechanism.
8. An interlocking cannula system comprising: a plurality of cannulas comprising a first cannula, a second cannula, and a third cannula, wherein the first cannula is smaller than the second cannula and the third cannula, and the second cannula is smaller than the third cannula, thereby allowing the first cannula to be nested within the second cannula and/or the third cannula, wherein each of the plurality of cannulas includes a respective locking mechanism, which allows the first cannula to be interlocked with the third cannula without the second cannula therebetween and which allows the second cannula to be interlocked with the third cannula, wherein each of the respective locking mechanisms includes a female locking component and a male locking component, wherein each respective cannula of the plurality of cannulas includes a head including a first side and a second side, and a tube extending from the second side of the head, and wherein the head includes the female locking component on the first side and the male locking component on the second side.
9. The interlocking cannula system of claim 8, wherein the male locking components of each of the plurality of cannulas are substantially identical.
10. The interlocking cannula system of claim 8, wherein the female locking components of each of the plurality of cannulas are substantially identical.
11. The interlocking cannula system of claim 8, wherein a cross-sectional shape of the male locking component comprises two straight sides and two curved sides.
12. The interlocking cannula system of claim 8, wherein the male locking component comprises an extension, wherein the female locking component comprises a notch configured to receive the extension, and wherein the notch comprises a circular cross-section which allows the extension to rotate about an axis when received in the notch.
13. The interlocking cannula system of claim 8, wherein the male locking component comprises an extension, wherein the female locking component comprises a notch configured to receive the extension, and wherein the notch comprises a rectangular cross-section.

14. An interlocking cannula system comprising: a plurality of cannulas comprising a first cannula, a second cannula, and a third cannula, wherein the first cannula is smaller than the second cannula and the third cannula, and the second cannula is smaller than the third cannula, thereby allowing the first cannula to be nested within the second cannula and/or the third cannula, wherein each of the plurality of cannulas includes a respective locking mechanism, which allows the first cannula to be interlocked with the third cannula without the second cannula therebetween and which allows the second cannula to be interlocked with the third cannula, wherein the respective locking mechanisms are configured such that interlocking one cannula of the plurality of cannulas with another cannula of the plurality of cannulas includes compressing the one cannula inward.

15. An interlocking cannula system comprising: a plurality of cannulas comprising a first cannula, a second cannula, and a third cannula, wherein the first cannula is smaller than the second cannula and the third cannula, and the second cannula is smaller than the third cannula, thereby allowing the first cannula to be nested within the second cannula and/or the third cannula, wherein each of the plurality of cannulas includes a respective locking mechanism, which allows the first cannula to be interlocked with the third cannula without the second cannula therebetween and which allows the second cannula to be interlocked with the third cannula, wherein the respective locking mechanism of the first cannula includes a first threading and a cam, wherein the respective locking mechanism of the third cannula includes a third threading and a first female locking component, wherein the respective locking mechanism of the second cannula includes a second threading and a second female locking component, wherein the second threading is configured to engage the first female locking component thereby interlocking the third cannula and the second cannula, and wherein the first threading is configured to engage the first female locking component thereby interlocking the first cannula and the third cannula, and wherein the cam is configured to engage the second female locking component thereby interlocking the second cannula and the first cannula.

16. The interlocking cannula system of claim 15, wherein the second threading has a larger thread pitch than the third threading.

17. The interlocking cannula system of claim 15, wherein the first female locking component is configured such that the cam cannot engage with the first female locking component.

18. An interlocking cannula system comprising: a plurality of cannulas comprising a first cannula, a second cannula, and a third cannula, wherein the first cannula is smaller than the second cannula and the third cannula, and the second cannula is smaller than the third cannula, thereby allowing the first cannula to be nested within the second cannula and/or the third cannula, wherein each of the plurality of cannulas includes a respective locking mechanism, which allows the first cannula to be interlocked with the third cannula without the second cannula therebetween and which allows the second cannula to be interlocked with the third cannula, wherein the respective locking mechanism of the first cannula includes a first threading and a first coil spring, wherein the respective locking mechanism of the third cannula includes a third threading and a first female locking component, wherein the respective locking mechanism of the second cannula includes a second threading, a second coil spring, and a second female locking component, wherein the second coil spring is configured to engage the first female locking component thereby interlocking the third cannula and the second cannula, and wherein the first coil spring is configured to engage the first female locking component thereby interlocking the first cannula and the third cannula, and wherein the first coil spring is configured to engage the second female locking component thereby interlocking the second cannula and the first cannula.

19. The interlocking cannula system of claim 18, wherein the second coil spring comprises a greater spring force than the first coil spring.
